2 A program, written using object-oriented programming, stores pictures as objects.

The program stores the dimensions of the picture (width and height), the colour of the frame (e.g. black), and a description of the picture (e.g. flowers).

The class has the following attributes and methods.

Picture	

`Description : STRING`
`Width : INTEGER`
`Height : INTEGER`
`FrameColour : STRING`
	`// stores a description of the picture`
	`// stores the width e.g. 30`
	`// stores the height e.g. 40`
	`// stores the colour e.g. black`

`Constructor()`
 	
`GetDescription()`
`GetHeight()`
`GetWidth()`
`GetColour()`
`SetDescription()`
	`// takes all four values as parameters and sets them to the private attributes`
	`// returns the description of the picture`
	`// returns the height`
	`// returns the width`
	`// returns the frame colour`
	`// takes the new description as a parameter and writes the value to description`

(a) The constructor takes the picture description, frame colour, height, and width as parameters and sets these to the private attributes.

Write the program code to declare the class `Picture` and its constructor.
Do not write any other methods.

Use your language appropriate constructor. All attributes should be private.

If you are writing in Python programming language, include attribute declarations using comments.

Save your program as **question 2**.

Copy and paste the program code into **part 2(a)** in the evidence document.

[5]

(b) The four get methods return the associated attribute, for example, `GetDescription()` returns the description of the picture.

Write the **four** get methods.

Save your program.

Copy and paste the program code into **part 2(b)** in the evidence document.

[3]

(c) The method `SetDescription()` takes a new description as a parameter, and writes this value to the appropriate attribute.

Write the method `SetDescription()`.

Save your program. Copy and paste the program code into **part 2(c)** in the evidence document.

[2]

(d) Write program code to declare an array of type `Picture` with 100 elements.

Save your program. Copy and paste the program code into **part 2(d)** in the evidence document.

[1]

(e) The text file `Pictures.txt` stores the data for the pictures in the order: description, width, height, colour.

For example, for the first picture in the text file:

`Flowers` is the description
`45` is the width
`50` is the height
`black` is the frame colour.

The data read into the program from the text file is stored in an array of type `Picture`.
The main program and the function will need to access the array data.

The function `ReadData()`:

- opens the file `Pictures.txt`
- reads the data from the file
- creates a new object of type `Picture` for each picture
- writes each object to the array
- raises an exception if the file cannot be found
- counts and returns the number of pictures in the array.

Write program code for the function `ReadData()`.

Save your program. Copy and paste the program code into **part 2(e)** in the evidence document.

[8]

(f) The main program calls the function `ReadData()`.

Write the main program.

Save your program. Copy and paste the program code into **part 2(f)** in the evidence document.

[2]

(g) The main program needs to ask the user to input their requirements for a picture. The user will enter the colour of the frame, the maximum width, and the maximum height of the picture.

The program will then search the array of pictures, and output the picture description, the width, and the height of any picture that meets the user's requirements.

The program should allow the user to input the colour in any case (e.g. Silver, silver, or SILVER), and still output the correct results.

Edit the main program to perform the described actions.

Save your program. Copy and paste the program code into **part 2(g)** in the evidence document.

[7]

(h) Test your program by inputting the following search criteria:

- BLACK, 100, 100
- silver, 25, 25

Take screenshots to show the output for both search criteria.

Copy and paste the screenshots into **part 2(h)** in the evidence document.

[2]